

CSE 4412: Introduction to Quantum Computing, Cryptography, and Networking.

Walter O. Krawec

Fall 2026

1 Instructor Information

- Instructor: Walter O. Krawec
- Instructor's Office: ITE 245
- E-mail: walter.krawec@uconn.edu

2 Class Meetings and Office Hours

- Class Meeting: Tu/Th 11:00am-12:15pm
- Classroom: ITE 127
- Instructor's Office Hours: Wednesdays 11:00am-noon; Thursdays 2:00-3:00pm; and by appointment
- TA Office Hours: Announced on Husky CT

3 Course Information

- Title: CSE 4412: Introduction to Quantum Computing, Cryptography, and Networking.
- About: Fundamentals of quantum computing (qubits, superposition, measurements, quantum circuits) including basic quantum algorithms and quantum complexity. Quantum cryptography and networking, including quantum key distribution protocols and their security proofs, and the design and use of large-scale quantum networks (e.g., quantum repeaters). Hands-on programming of the IBM quantum computer.
- Prerequisites: CSE 3500 or 5050; MATH 2210Q.

- Recommended Reading:
 1. "Quantum Communication and Cryptography" by Walter O. Krawec. Pre-print chapters available on HuskyCT as needed.
 2. "Quantum Computer Science" by N. David Mermin. ISBN 978-0-521-87658-2.
 3. "Quantum Computation And Quantum Information" by Michael A. Nielsen and Isaac L. Chuang
 4. http://web.cecs.pdx.edu/~mperkows/CLASS_FUTURE/NEW_MATERIALS_2011/lukac_perkowski_book_introduction_and_quantum_mechanics.pdf
 5. <https://batistalab.com/classes/v572/Mosca.pdf>
- Course Outcomes:
 1. Understand the main postulates of quantum computing, including state vectors, measurements, and unitary evolution.
 2. Learn how to analyze quantum algorithms, including Deutsch–Jozsa, Bernstein–Vazirani, Shor’s, and Grover’s.
 3. Discover how quantum information can be used for cryptography, focusing on quantum key distribution.
 4. Investigate practical concerns surrounding quantum cryptography and communication, focusing on quantum networking approaches.

4 Course Grading

- Homework: 35%
 - Several take-home assignments will be given throughout the semester. Unless otherwise specified by the instructor, you may work in groups of 2-3 on these assignments. If you choose to work in groups, all members of that group should have their names clearly printed on the first page of the assignment. Furthermore, all members of the group must submit the same document (see below for hand-in policy). Group members will be given the same grade for the assignment.
 - Several programming assignments will also be given. These will involve working with the IBM quantum computer and simulator in "Qiskit"
 - All take-home assignments will be announced on Husky CT
- Exams: 60%
 - There will be three in-class exams, each worth 20%
 - More information on the topics covered on the exam will be provided as the test day approaches

- Reverse Homework: 5%
 - Once a week, you must submit a "reverse HW" to the instructor, which asks one question about something covered **in class lecture that week**.
 - I will not answer all questions, but some I may choose to answer in class.
 - These are due **Sunday** of every week on HuskyCT.
 - Only a sentence or two suffices! No more than a paragraph please (any longer question you should ask in class or office hours)!
- Note: Extra credit assignments are sometimes given out during lectures. If you miss a lecture, you are responsible for finding out if any such assignment was given, as these extra credit assignments may not be announced on Husky CT.

5 Submission Policy:

All written assignments must be submitted electronically to HuskyCT as PDF files (no other format will be accepted!). They should either be typed and submitted as a PDF file, or scanned and submitted (photos of assignments will not be accepted – use a real scanner or a very good camera scanning app!). If you scan your document, make sure to check it is legible. The instructor reserves the right to give a 0 to any answer that is not legible.

If you work in a group, all members of that group should have their names listed clearly on the assignment hand-in. Furthermore, all members of the group should upload the same submission to HuskyCT. All members of the group will be given the same grade.

While AI use is allowed for Homework assignments, it is not recommended. **If you are stuck on a problem, come and see the instructor during office hours, or ask in class!** AI use, or any other technological development beyond paper and either pencil or pen, will not be allowed for exams.